ELECTRIC VEHICLE MANAGEMMENT SYSTEM USING ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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Project Proposal Report

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DECLARATION

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Electric vehicles (EVs) emerge as another critical solution for sustainable transportation, as they can reduce greenhouse gas emissions and reliance on fossil fuels. However, the issue of battery degradation is one of the biggest barriers to EV adoption. The capacity of lithium-ion batteries in EVs deteriorates over time due to frequent charging, high temperatures, deep discharge, and aggressive driving. This degradation plays a significant role in reducing driving range and increasing maintenance costs, while also minimizing overall battery life.

Current EV battery monitoring systems fail to provide information other than battery percentage and real-time state of charge. Such systems lack predictive features that can predict future battery health based on user behavior. Also, most of the current literature is based on laboratory data and not on real-life driving and charging trends, limiting the opportunities to provide individual analysis to EV users.

This study aims to present a battery health prediction system that is adaptive and adopts machine learning to interpret the behavior of the battery to predict the state of health (SOH). The parameters analyzed in the system include charging frequency, charging time, battery temperature, acceleration pattern, travel distance and energy consumption. The prediction of EV battery health is performed using machine learning models created in Python, Scikit-learn, TensorFlow, and data visualization frameworks to provide personalized recommendations to EV users.

The projected outcome of the given study is a smart predictive system that improves battery life, increases energy efficiency, and helps EV users develop more favorable charging and driving behaviors. The system can also be beneficial to EV manufacturers and service providers by providing proactive battery management and reducing battery replacement costs.

Keywords Electric Vehicles, Battery Health Prediction, Machine Learning, Charging Behavior, Driving Behavior.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
EV	Electric Vehicle
BMS	Battery Management System
DoD	Depth of Discharge
IEA	International Energy Agency
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
RMSE	Root Mean Square Error
RBAC	Role-Based Access Control
RUL	Remaining Useful Life
SDG	Sustainable Development Goal
SOC	State of Charge
SOH	State of Health
XGBoost	Extreme Gradient Boosting
OBD	On-Board Diagnostics
OEM	Original Equipment Manufacturer
PII	Personally Identifiable Information
SEI	Solid Electrolyte Interphase

1. INTRODUCTION

1.1. Background and Context

Electric vehicles (EVs) have emerged as one of the promising solutions to reduce greenhouse gas emissions and the environmental threat posed by conventional fuel-powered vehicles. As the world becomes increasingly concerned about the issue of climate change and the sustainability of energy resources, the use of EVs is gaining popularity among various governments and organizations due to their clean energy solutions. However, the battery system is one of the most important components that precondition the performance, reliability, and cost of an electric vehicle.

Lithium-ion batteries are the other most used battery types in EVs due to their efficiency and high energy density. Despite these benefits, a major problem of battery degradation remains. Over time, EV batteries have a capacity that decreases with each charging cycle, temperature, aggressive driving habits, and deep discharge characteristics. This degradation has a direct impact on the vehicle's driving range, its performance, and its overall lifespan. As a result, EV users tend to experience some unexpected losses in battery efficiency, which can lead to costly battery changes.

Battery degradation impacts various stakeholders such as EV owners, manufacturers, maintenance service providers, and energy infrastructure operators. EV users are likely to experience reduced driving range and higher maintenance costs, while manufacturers must address reliability issues in terms of battery performance. Also, battery replacements when battery life is not well-organized lead to electronic waste and environmental impacts.

The International Energy Agency (IEA) says that there will be more than 26 million electric vehicles worldwide in 2022, and the EV market is expanding rapidly every year. Electric mobility in Sri Lanka is also gaining momentum due to high fuel prices and concerns over environmental issues. As the number of EVs grows, the battery health management process is becoming more important.

Most current EV monitoring systems do not provide more than basic information, including battery percentage, state of charge, and projected driving range. These systems fail to provide any predictive experience that can help their users understand how their driving patterns and battery charging will affect battery health in the long term. Moreover, most battery management systems use sensor-based information and laboratory tests but do not examine actual real-world behavior patterns.

This study aims to address such shortcomings by proposing a framework for an adaptive battery health prediction system to predict battery degradation trends by examining user charging and driving habits. In the proposed system, machine learning algorithms are applied to examine parameters such as charging frequency, charging time, battery temperature, acceleration, trip range, and energy usage. The system will be able to predict future degradation trends by determining patterns of user behavior to estimate the battery state of health (SOH).

Various important components such as a data collection module, data preprocessing, machine learning prediction engine, and a recommendation system are proposed in the system. The components are collectively used to provide accurate battery health predictions and recommendations for EV users to act.

This study can be attributed to both the United Nations Sustainable Development Goal (SDG) 7 - Affordable and Clean Energy and SDG 13 - Climate Action, as the research will ensure that battery life is improved to reduce environmental impact and energy waste. Furthermore, this study belongs to the Artificial Intelligence and Data Science research cluster aimed at predictive analytics and effective decision-support systems in sustainable technologies.

Figure 1.1 provides a visual representation to better understand the overall concept of the proposed system. This figure shows the most important elements and relationships of the adaptive battery health prediction framework. It also shows the relationship between electric vehicle development, battery degradation issues, and the limitations

of current monitoring and how machine learning methods can be applied to improve battery health prediction and sustainability.



Figure 1.1: Conceptual Overview of Battery Health Prediction System

The proposed system consists of a combination of various components, as shown in Figure 1.1, which include data collection, machine learning models, and recommendation mechanisms to determine battery health based on charging behavior and driving behavior. The diagram also highlights critical issues such as capacity loss and lack of predictive capabilities in current systems and links the solution to EV development and sustainability on a global scale. This general outline illustrates that smart analytics can help ensure improved battery management and, in addition, improve the lifespan and performance of electric vehicles.

1.2. Literature Review

Electric vehicles (EVs) are gaining popularity around the world fast, and this factor has increased the significance of reliable and smart battery health management systems. Lithium-ion batteries have become the main storage of energy used in EVs because they have a high energy density, a comparatively low rate of self-discharge, and a decreasing cost of production. These batteries, however, are prone to progressive degradation due to electrochemical reactions in their cells, thereby lowering the vehicle driving range, raising the cost of maintenance, and reducing the operational life of the vehicle. With over 26 million EVs on the road in the world in 2022, according to the International Energy Agency [1], the rapidity of making complex battery health monitoring and prediction systems has become more evident than ever. A predictive battery health system that uses machine learning to examine real-world user charging and driving behavior is a very timely and needed contribution to this field.

The suggested system incorporates a data-gathering component, a pre-processing pipeline, a machine learning prediction engine, and a recommendation system accessible to the user in a single framework. The desired future functionality of the system involves forecasting battery State of Health (SOH) degradation behavior based on the idea of learning through individual user behavior patterns, such as charging frequency, charging duration, battery temperature, acceleration intensity, trip range, and energy consumption, and converting these predictions into practical advice to enable EV users to maintain the battery life. The next sections critically review the recent academic literature on the battery degradation mechanisms, SOH estimation techniques, available battery management products and services, and constraints that justify the proposed system.

1.2.1. Existing Research Approaches

The study on lithium-ion battery degradation and SOH estimation has developed significantly during the last ten years. Edge et al. [2] offered a detailed analysis of the degradation processes, dividing them into calendar aging, which is a process based on the storage temperature and the state of charge, and cycle aging, which is a process of repeated charge and discharge. Electrochemical processes of importance are the development of the solid electrolyte interphase (SEI) layer, lithium plating, and structural disordering of electrode materials, which are reflected in capacity loss and internal resistance, which increase over time. This basic knowledge of the degradation processes forms the basis of the strategy of feature selection of the proposed framework.

Data-oriented methods have become of special concern in place of computationally intensive electrochemical and equivalent circuit models. Severson et al. [3] also showed that early-cycle charging data was predictive of long-term battery cycle life before these systems started to show decreased capacity, which set a precedent of using operational data to predict subsequent health patterns. Roman et al. [4] took the next step in this direction and introduced a temperature-conscious machine learning pipeline to estimate the SOH by confirming the importance of thermal parameters to the accuracy of the predictions. Based on these results, Lu et al. [5] created a deep learning model that is trained on data of 65 commercial batteries in five manufacturing companies and that, with a domain-adaptation strategy, results in SOH estimation errors of less than 3 percent on 89.4 percent of test samples.

Long Short-Term Memory (LSTM) networks have proven to be a rather useful architectural choice for battery health prediction because of their ability to capture long-range temporal dependencies in sequential data of charging and discharging. Jorge et al. [6] suggested an LSTM-based multi-step-ahead SOH forecasting system, which forecasts the degradation behavior in the future using a sliding window of historical BMS data, which proved that it is possible to make predictions of future health based on the real-world operation history. Hybrid models have also performed well, e.g., Krishnasamy and Alagar Muniandi [7] presented a hybrid RF-LSTM-XGBoost that has lowered the RMSE and MAE by 7-12 percent on NASA batteries

compared to the traditional LSTM baselines, indicating that integrating feature selection, time modeling, and gradient boosting into a single prediction pipeline is beneficial.

Another direction of development of research is physics-informed hybrid models. Amiri et al. [8] suggest combining physics-based electrochemical models with machine learning to restrict the solution space and enhance extrapolation to novel operating conditions, whereas Singh et al. [9] assessed a physics-informed neural network in battery state estimation and showed enhanced performance compared to data-driven models only. The hybrid approaches provide the benefits of interpretability and maintain the flexibility of machine learning, which is why they are suitable to be deployed to intelligent BMS systems.

The contribution of real-world driving and charging behavior to battery aging is also getting a growing body of research. Jiang et al. [10] showed that behavior-directed monitoring of battery health based on extreme learning machines can be used to predict SOH based on the speed profiles, acceleration events, and braking intensity information of a real vehicle. Wang et al. [11], which examined data on the operations of 300 various EVs with three years of measurement, proved that large-scale real-world data can significantly enhance the accuracy of SOH prediction compared to those based on the models determined in the laboratory. All these studies add to the idea that user behavioral data has a powerful predictive value in terms of battery health and must be methodically introduced into the prediction systems.

1.2.2 Existing Products and Services

There are several commercial battery management solutions and services that are currently offered in EV battery monitoring. All the modern EVs include standard OEM battery management systems, which have been reviewed by Liu et al. [12] to offer real-time cell voltage, current, temperature and state of charge. These systems have critical safety tasks like overcharge protection, thermal control, and cell balancing. Their SOH estimation abilities are however normally restricted to simple Coulomb counting or empirical look up tables which are not adjustable to the individual user behavior.
Third-party diagnostic applications like Lectron, Spritmonitor and other cloud-based telematics services give EV owners a better battery health report, including past count of charge cycles, capacity estimates, and range predictions. These services combine data on the OBD-II port or manufacturer API of the vehicle and display them in mobile apps. These tools provide a higher level of visibility than factory BMS interfaces but are more retrospective than predictive (they report what has already happened and not what to recommend changing behavior and reduce future degradation) and these tools do not provide personalized advice.

Prototypes of scholarship BMS reviewed by Habib et al. [13] use more advanced estimation algorithms, such as Extended Kalman filter (EKF) and variations of Unscented Kalman filter (UKF) algorithms, to provide better SOC and SOH tracking. Fleet management systems based on cloud computing offered by Kumar et al. [14] have already started to investigate the possibilities of aggregating multi-vehicle data to enhance the accuracy of the model and provide an opportunity to plan maintenance beforehand. Nevertheless, none of the commercially available systems have combined customized user behavior analysis, machine learning-trained SOH prediction, and a recommendation engine in the single, adaptable framework available to individual EV users.

1.2.3. Limitations of Existing Systems

An analysis of literature and business systems shows that there are several inherent constraints. To begin with, the overwhelming use of laboratory data (e.g., the NASA PCoE battery dataset and CALCE dataset) implies that most published models are trained in idealized, standardized cycling simulated conditions that are not representative of the diversity of realistic EV usage. The results of Wang et al. [11] and Mathieu et al. [15] have shown that there is a large discrepancy between the laboratory-predicted and field degradation patterns, and this implies that the models that are trained on the laboratory data only show poorer accuracy and generalization when applied in the real car.

Second, the current systems offer little personalization. The commercial BMS platforms and third-party diagnostic tools display aggregate health indicators according to population means or predetermined degradation curves, instead of being adjusted to the individual charging behavior, driving style, or environmental conditions. Selvararaj and Vairavasundaram [16] stated that EV batteries under real-world conditions experience non-static load profiles that cannot be effectively captured by a static model, and instead this paper recommends adaptive, real-time estimation frameworks.

Third, and most importantly, there is no system available that would bridge the gap between prediction of battery health and recommendations that users could act on. In a thorough review, Ahmad et al. [17] validated the fact that although predictive accuracy of SOH models has been significantly enhanced, practically none of the systems can convert the degradation predictions into physical guidance to allow ordinary users to change their habits and increase battery life. Rauf et al. [18] also observed that feature selection, that is, which behavioral parameters correlate best with degradation, has not been systematically applied to personalized and user-specific BMS settings. Bhattacharya [19], on the same note, found out that real-time flexibility to various and changing driving patterns is one of the unresolved issues of present BMS systems. All these gaps indicate the necessity of the adaptive, behavior-aware, and recommendation-driven framework that is suggested in the current work.

1.3. Research Gap

Despite significant advances in predicting the state of health (SOH) of lithium-ion batteries using machine learning, deep learning, and hybrid models, the existing literature has largely focused on improving prediction accuracy in controlled settings rather than creating deployable, user-friendly intelligent systems. The majority of existing methods use standardized laboratory data and fixed pipeline training models that do not reflect the reality of the electric vehicle usage problem, which is complex, variable, and unique.

The main weakness of literature is the lack of long-term and multi-horizon prediction models that can predict real-world behavioral data with continuously changing degradation trends (3, 6, 12 months). Moreover, while a body of literature acknowledges the impact of driving behavior, charging patterns, and temperature on the battery degradation process, there is no formal attempt to integrate them into an adaptive and personalized modeling framework.

Moreover, current models are typically standalone prediction engines, lacking explanatory mechanisms or recommendation layers that can translate technical predictions into practical advice for their end users. The failure to integrate data collection, adaptive learning, long-term forecasting, and decision support results in disjointed research applications that cannot be readily adapted as complete battery health management packages.

Therefore, the inherent research gap consists in the fact that an integrated, adaptive, behavior-based, multi-horizon battery health prediction and recommendation framework has not been developed and can be implemented on real-world operational data within a single end-to-end system architecture.

1.4. Research Problem

The high rate of electric vehicle globalization has increased the pressure on lithium-ion batteries as the main energy storage technology. Although current battery management systems (BMS) include all the necessary monitoring capabilities, including state of charge (SOC) and safety features, they lack long-term predictive intelligence as well as customized decision support for specific users.

Without integrated and adaptive predictive systems, EV users cannot understand the impact of their specific driving practices, charging schedules, and environmental temperature conditions on battery degradation in the future. Existing solutions only provide short-term health measures without predicting the impact of degradation over longer time periods such as 3, 6, and 12 months. As a result, users cannot take proactive steps to maintain the health of their batteries before serious degradation is seen.

Moreover, where SOH prediction models exist, their results are often in the form of numerical values of health without interpretable explanations and actionable suggestions. This creates an imbalance between technical prediction and practical utility so that a user is unable to adapt his behavior in a way that maximizes battery life.

Thus, there is a strong need to have a smart, dynamic, and user-friendly battery health management system that:

- Predicts long-term SOH degradation trends
- Integrates real-life battery, behavioral, and environmental data
- Adapts to individual use
- Provides practical and customized suggestions to maximize battery performance and capitalization

2. OBJECTIVES

2.1. Main Objectives

Developing a machine learning-powered system to predict electric vehicle (EV) battery degradation after 3, 6, and 12 months using real-world battery data, driving behavior, charging behavior, and temperature data, and providing personalized advice to assist users in preserving and extending battery life.

2.2. Specific Objectives

1). To determine and incorporate multi-source real-world information pertaining to battery degradation.

This goal aims at a systematic gathering and preprocessing of dissimilar data information, with battery working parameters (ex: state of charge, depth of discharge, voltage, current, number of cycles), charging behavior measurements (ex: charging frequency, charging rate, charging duration), driving behavior measurements (ex: average speed, acceleration patterns, trip frequency, idle time), and environmental temperature measurements. The combination of these variables is to build a complete feature representation to have accurate degradation modeling.

2). Create predictive long-term battery health machine learning models.

This work aims to design, train, and optimize machine learning and deep learning algorithms suitable for assessing battery degradation behavior and predicting future SOH values. The system will be based on multi-horizon forecasting, where it will produce forecasts for 3 months, 6 months, and 12 months to be used in short-term monitoring as well as long-term strategic decision-making.

3). Compare and contrast different modeling strategies to find the most suitable forecasting model.

This experiment will be conducted to evaluate different forecasting methods through appropriate performance evaluation measures. Its aim is to identify the most accurate, robust, and computationally efficient model that can represent the nonlinear interdependence of behavioral variables and battery degradation curves.

4). Predict the dependence between user behavior and environmental conditions and battery degradation trends.

The aim of this goal is to evaluate the impact of charging habits, driving behavior, and temperature changes on SOH degradation. Learning behavior-based battery aging processes and analyzing them in terms of more interpretable and explainable predictive results will be important.

5). Design and implement a customized recommendation system to optimize battery life.

The system will produce personalized data-driven recommendations based on assumptions about SOH patterns and behavior analysis to reduce unnecessary stress on the battery. These recommendations may include strategies to optimize charging, driving, and preventive maintenance to help increase battery life and contribute to sustainable energy use.

3. METHODOLOGY

3.1. Research Approach

In this study, a quantitative methodology with the help of machine learning developed an intelligent battery State of Health (SOH) long-term prediction system for Electric Vehicles (EVs). The research has a well-organized pipeline, which includes multi-source data collection, data preprocessing, feature extraction, predictive modeling, and recommendation generation. The method combines actual driving behavior, charging behavior, battery operation data, and environmental temperature to develop an adaptive and customized battery degradation forecasting model. Figure 3.1 shows the general system architecture of the proposed solution.

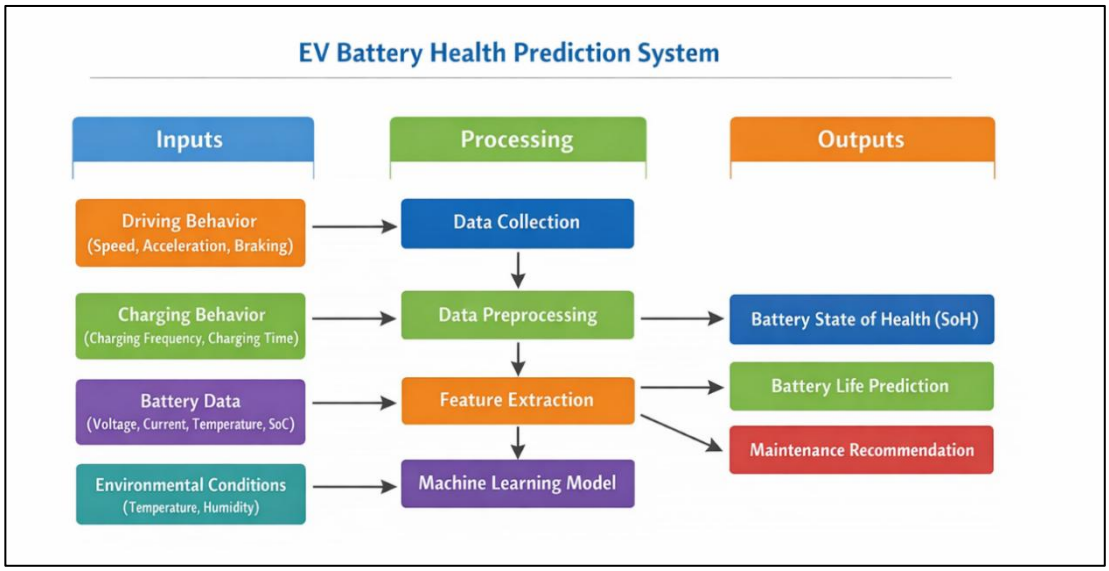


Figure 3.1-Conceptual Framework of Behavior-Based Battery Health Prediction

Figure 3.1 provides a high-level architecture of the proposed EV battery health prediction system. The system is organized into three main steps, which include input, processing, and outputs.

Driving behavior data, charging behavior data, battery operating parameters, and environmental conditions are the input stages. These data are processed in the flow processing stage and are systematically collected and processed in the processing stage, which includes data preprocessing, feature extraction, and machine learning model

building. The trained model produces three outputs for battery state of health (SOH) prediction, long-term battery life prediction (3, 6, and 12 months), and customized maintenance suggestions. This is a mechanistic process that ensures that raw real-world data is transformed into useful predictive information and actionable decisions regarding battery health optimization.

3.2 Technical Methods and Algorithms

The proposed research will use a data-driven machine learning approach to predict and model battery state of health (SOH) degradation based on multi-source real-world information. The system has a systematic pipeline that includes data collection, preprocessing, feature engineering, model development, and prediction generation.

First, raw data collected on battery systems, charging behavior, driving patterns, and environmental conditions are preprocessed to remove noise, missing values, and feature normalization. Feature extraction methods are used to obtain useful indicators such as depth of discharge (DoD), charging frequency, trip intensity, and temperature changes, which play a critical role in analyzing battery degradation.

Long Short-Term Memory (LSTM) networks are often used in predictive modeling due to their ability to capture time-dependent relationships and sequential patterns in time-series data. LSTMs are particularly useful in modeling trends in battery degradation over time and generating multi-horizon forecasts (3, 6, and 12 months). In addition, standard machine learning models such as Random Forest and XGBoost can be implemented and tested to compare them for reliability and accuracy.

The predicted values of SOH and battery life trends are the final model outputs, which are then used to produce a set of personalized recommendations using a rule-based or data-driven recommendation module. This combination ensures proper prediction while facilitating effective decision-making.

3.3. Tools and Platforms with Justification

The development of the proposed system relies on a combination of programming tools, machine learning frameworks, and application development platforms. The selected tools and their justifications are summarized in Table 3.1.

Table 3.1: Tools and Platforms with Justification

Tool / Technology	Purpose	Justification
Python	Backend development & ML modeling	Widely used for data science, robust libraries (TensorFlow, Pandas, Scikit-Learn).
LSTM (Deep Learning)	SOH prediction	Time series effectively capture patterns and long-term dependencies.
Random Forest / XGBoost	Model comparison	Provides baseline and improves forecast robustness
React Native	Frontend mobile app	Enables cross-platform (Android and iOS) development
Firebase	Database & backend services	Real-time data storage, scalability, and secure cloud integration
TensorFlow / Keras	ML framework	Efficient model building and training for deep learning
Google Colab / Jupyter Notebook	Development environment	Easy experiments and visualization
GitHub	Version control	Supports collaboration and code management

This table provides the important technologies used in the system, their roles, and justifications. The choice of tools is driven by their scalability, performance, ease of development, and compatibility with machine learning workflows. Python and TensorFlow will allow for successful model building, while React Native and Firebase will help launch a user-friendly and real-time application without any issues.

3.5. Validation Strategy and Performance Metrics

A detailed validation plan is implemented to make the proposed system reliable, accurate, and user-friendly. The analysis is based on the performance of the technical model and the efficiency at the user level. The performance metrics used to evaluate the system are summarized in Table 3.2

Table 3.2: Performance Metrics and Evaluation Criteria

Metric	Description	Target
Accuracy	Measures the overall accuracy of predictions.	≥ 85%
F1-Score	Balances accuracy and memory	≥ 0.85
RMSE (Root Mean Square Error)	Measuring the magnitude of forecast errors	Minimized
MAE (Mean Absolute Error)	Average absolute forecast error	Minimized
Latency	Time taken to generate predictions	≤ 10 seconds
User Satisfaction	User feedback on the use of the system	≥ 80%

The main performance indicators to be applied in assessing the effectiveness of the proposed system are shown in this table. Classification or prediction quality is measured by accuracy and F1-score, numerical prediction errors can be measured by RMSE and MAE. Latency ensures that the system is responsive to the user, and user satisfaction measures how practical the application is to use. Combined, these measurements provide a holistic assessment of technical performance and user experience.

Validation Strategy

The proposed system is tested in several ways to validate it to achieve the required robustness and generalizability:

- **Train-test split/cross-validation:** The dataset is split into training and test groups, or k-fold cross-validation is used to prevent overfitting and ensure model reliability.
- **Comparative model evaluation with other models:** LSTM is compared with other models such as Random Forest and XGBoost to determine the best approach.
- **Real-world scenario testing:** For practical applicability, the system is tested with real or semi-real data.
- **User evaluation:** All users are asked about their usage and efficiency of the recommendations.

Data Sources

The system uses a variety of data sources to be diverse and realistic:

- Public EV battery datasets
- Datasets from open-source research.
- Artificial or semi-real data on charging and driving behavior.
- Applications (in the real world) (where possible).

No individual or sensitive user information is collected, and it will not violate data privacy and ethical behavior.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Component

The proposed electric vehicle intelligent decision support system is a system consisting of several inseparable analytical components that work together to enable smart after-sales EV management. Such features include battery health prediction, vehicle problem detection, charging support, and EV vendor performance analytics.

The part of the research dedicated to predicting long-term battery degradation with references to user charging patterns and driving behavior is called adaptive battery health prediction. The component contains machine learning models, which are expected to determine the impact of daily driving on the health of an EV battery over time. Figure 4.1 below gives a picture of how this component is integrated into the larger system.

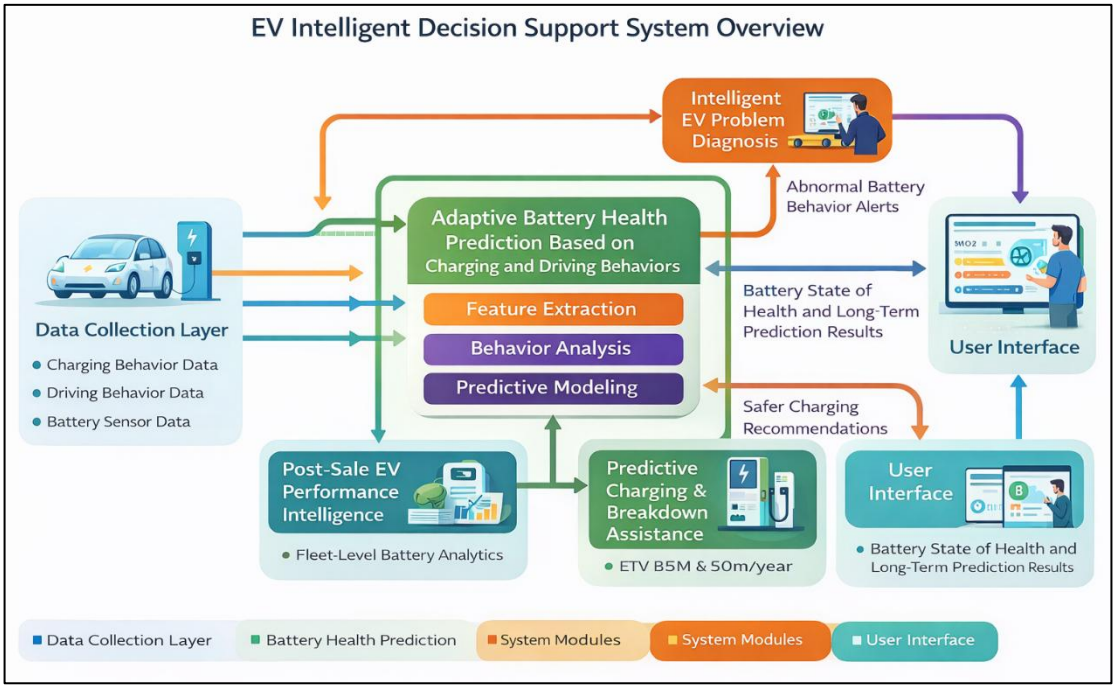


Figure 4.1: Overview of the EV Intelligent Decision Support System

Behavioral aspects that have a strong impact on battery degradation in electric vehicles are frequent fast charging, aggressive driving acceleration, high driving speeds, and high trips. Traditional EV dashboards typically only provide the current battery charge

level, which fails to provide an indication of the long-term health of their batteries. The proposed system overcomes this shortcoming by continuously monitoring behavioral information and predicting future battery health after 3, 6, and 12 months.

The component receives behavioral and sensor data as input at the EV data collection layer and processes them using a machine learning pipeline that includes processing components such as feature extraction, behavioral analysis, and predictive modeling. The output of the estimated battery health is then input to other modules of the system and the user interface and displayed to the user as a means of visualization and decision-making.

Also, this component interacts with other system modules. Examples of such applications include predictive battery degradation data to assist the intelligent EV problem diagnosis module in detecting abnormal battery behavior and predictive charging and breakdown support modules to suggest safe charging protocols. Furthermore, the collection of anonymous battery health trends can be communicated to the aftermarket EV performance module, allowing manufacturers to know what is happening with the battery in practice. Therefore, the high-level system architecture provides a scalable system that can integrate behavioral data, predictive analysis, and intelligent recommendations to improve the EV battery life cycle and support the decision-making process of users and EV stakeholders.

4.2 System Architecture Components

Adaptive Battery Health Prediction Module - This module consists of several interconnected operational layers to transform raw EV-related data into clear and useful insights. These parts of the system ensure a seamless flow from data collection to smart decision support.

4.2.1. Data Acquisition Layer

The data acquisition layer collects real-time and historical data related to EVs from various sources. It includes charging patterns such as fast and slow charging cycles and duration of charging sessions and driving patterns such as average speed, acceleration rate, and distance travel. It also includes vehicle usage patterns such as trip frequency, vehicle idle time, and energy consumption patterns.

It also includes battery health, including temperature, voltage level, and number of charging cycles. It collects data from the EV telemetry system, EV mobile app, and publicly available or simulated data.

This layer is essential for collecting consistent and reliable data in a well-structured manner and acts as the backbone for subsequent analysis.

4.2.2. Data Preprocessing Layer

The data preprocessing layer is the first stop on our machine learning journey. Here we take the raw data we've collected and clean it up to make it good and relevant. The world of electric vehicle data can often be noisy, missing, or messy, so this is about improving it and understanding what we have.

So, what's going on here? Well, some essential activities include cleaning the data, discarding bad records, filling in missing values, and normalizing all features so that their scale or range is consistent. Then there's the process of extracting meaningful signals from the raw data streams. This is advanced feature extraction. For example, charging logs and driving records are transformed into meaningful features like how many fast charges occur each week, how many long trips are taken, how aggressively the vehicle accelerates from standstill, and how much energy is used each day.

Then there's slicing and dicing the time series data so we can see how things change over time. It's all about helping the model understand temporal patterns.

4.2.3. Machine Learning Predictive Engine

The heart of the system is the machine learning prediction engine, which is a module that correlates how people drive with how their batteries degrade. It uses advanced algorithms such as random forest regression, sequential boosting, and LSTM networks to process the time-related data we collect.

The model is trained in historical data related to electric vehicle battery degradation and user behavior. Once trained, it is deployed and uses new incoming data to produce a prediction related to the health of the battery.

The model produces predictions related to different time horizons. These are short-term (approximately 3 months), medium-term (approximately 6 months), and long-term (approximately 12 months). These predictions give users a futuristic view of how their behavior today will affect their batteries in the future.

4.2.4. Insight Generation and Visualization Module

The Insight Generation and Visualization Module transforms raw forecast output into easy-to-understand insights. It does not display raw numbers but displays simple and clear messages that support decision-making.

It generates battery health trend charts, battery gauges of the rate of battery degradation, and the impact of your habits on battery health, e.g., the impact of fast charging habits on battery health. It also includes battery health maintenance tips.

It has an easy-to-understand interface via the mobile app, ensuring that the insights generated are easy to understand for the end user but can be shared with the rest of the data system.

4.2.5. Integration Layer with the Other System Components

The focus of the study is on battery health prediction, but the system is designed to be integrated with a larger smart EV decision support system. The role of the integration layer is to act as a bridge between this module and others.

The aftermarket EV performance intelligence system uses predictions to improve analytics and provide manufacturers with greater insight. They are also used by the EV problem diagnosis system to identify abnormal battery performance and potential battery problems. Furthermore, the predictive charging and breakdown support system uses predictions to provide the best charging strategy.

In this way, predictions are not used in isolation but are integrated with a larger smart EV management system that can meet the needs of users, service providers, and manufacturers.

4.3 High Level System Architecture Diagram

The high-level architecture of the adaptive battery health prediction system is structured into four different layers that work together to provide a valid and practical battery health prediction. The general architectural design of the proposed system is shown in Figure 4.2 below.

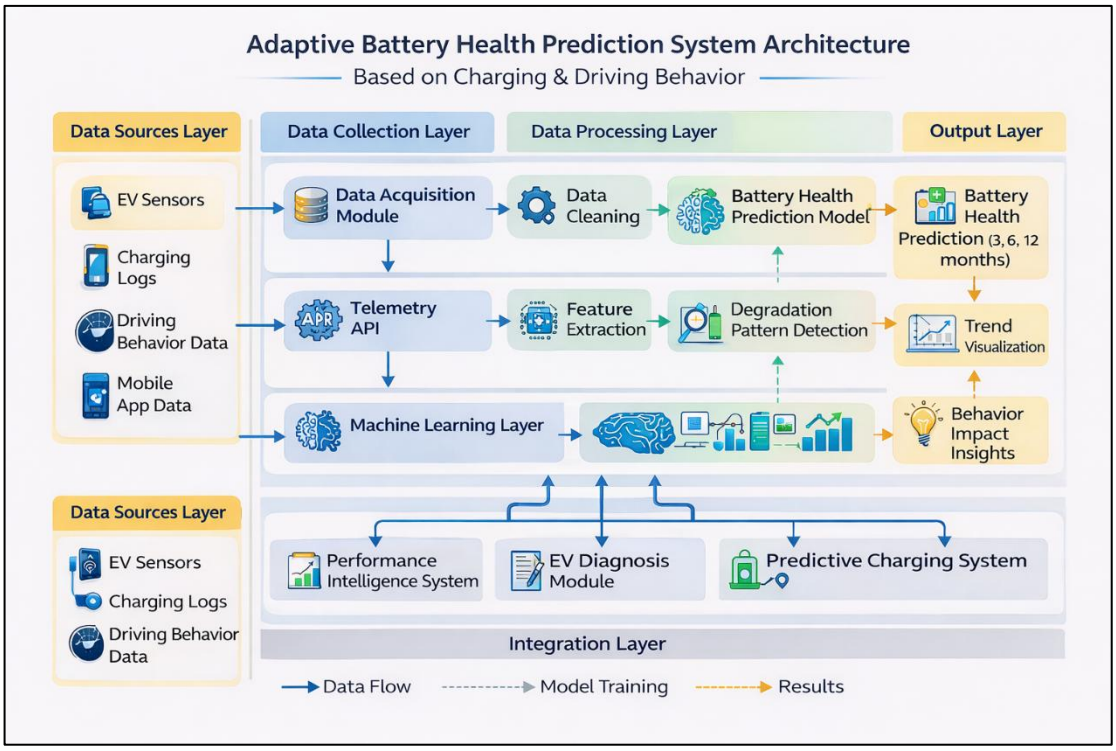


Figure 4.2: High Level Architecture of the Adaptive Battery Health Prediction System

The architecture is organized into four main layers as shown in Figure 4.2. The data source layer is the main gateway, which receives inputs from EV sensors, charging records, driving behavior data, and mobile application data. These are forwarded to the data collection layer, where the data acquisition module and telemetry API are used to acquire and integrate the raw data of the operation. The data processing layer then uses data cleaning, feature extraction, and machine learning algorithms to refine the raw inputs into meaningful patterns and feed them into the battery health prediction model and the degradation pattern recognition engine. The output layer provides results in three formats: battery health predictions, trend visualizations, and behavioral impact insights over 3-, 6-, and 12-month horizons. Furthermore, to ensure that battery health analytics are part of the overall intelligent EV management system, the bottom of the architecture is integrated with an integration layer that connects the system to the performance intelligence system, EV diagnostic module, and predictive charging system.

4.4 Component interactions

The components interact with each other in a sequential manner during the data processing flow. First, EV operational data is collected using the data acquisition module. This module is fed data from vehicle sensors, charging records, and user driving behavior records.

After the data collection step, the data preprocessing module is responsible for preprocessing the collected data. Feature extraction algorithms are used to transform the collected behavioral data into a structured model that can be used by machine learning algorithms.

After data preprocessing, the prediction engine is the next component in the sequence, and the machine learning algorithm uses patterns from the collected behavioral data to make predictions about the future state of the battery.

After the prediction engine makes predictions, the next step is to visualize the collected data using the visualization module, which generates trend charts and battery health indicators for the end user.

Finally, predicted battery health information is made available to other components of the intelligent system, enabling the operation of advanced analytics across the entire system.

4.5 Data flow explanation

The adaptive battery health prediction system depicts the path through which data travels from the external EV vehicle system through processing and finally to the result. The data flow diagram of the overall process of the proposed component is shown in Figure 4.3.

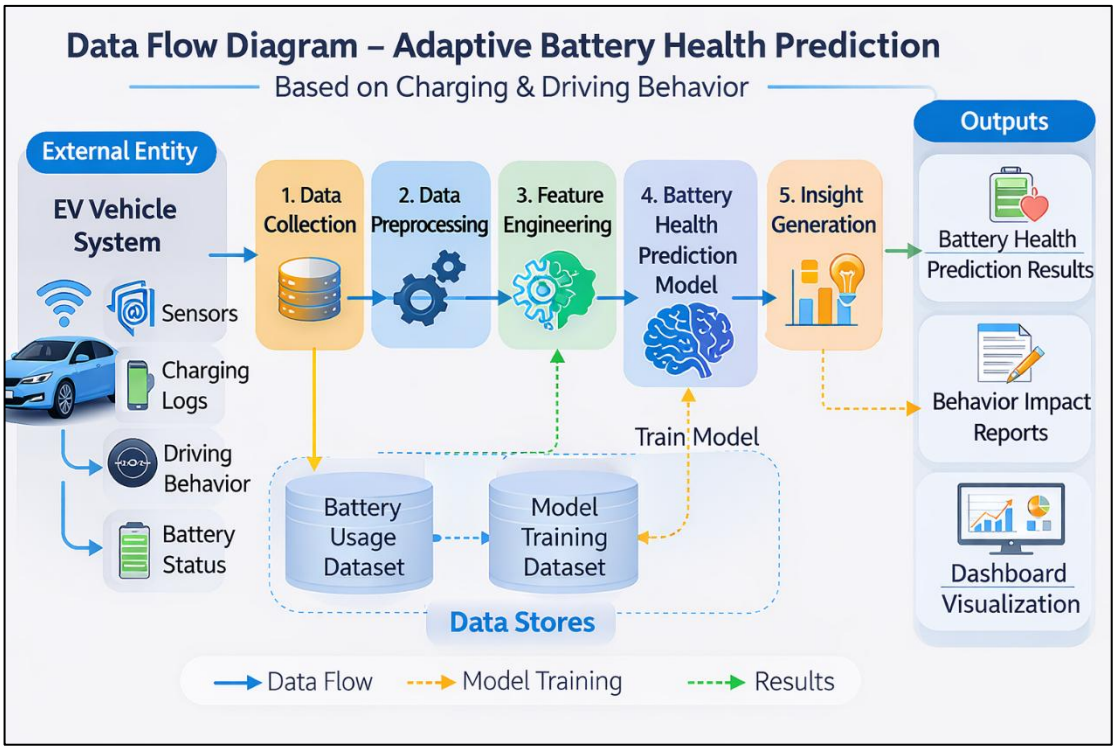


Figure 4.3: Data Flow of the Battery Health Prediction Component

In the diagram, we can see that the data flow starts from the EV vehicle system, which is our external source of continuous information input. Sensors, charging logs, driving behavior, and battery status are continuously input into the system. This data first enters Phase 1: Data Collection, where the collected data is stored in the battery usage dataset.

From here, Phase 2: Data Preprocessing refines the data by removing noise and inaccuracies. Then, Phase 3: Feature Engineering builds meaningful behavioral variables from the preprocessed data, creating the model training dataset.

This leads to Phase 4: Battery Health Prediction Model, where the model will be used to make predictions from the data. Finally, Phase 5: Insight Generation takes the forecast results and transforms them into usable formats such as battery health forecast results, behavioral impact reports, and dashboard visualizations, empowering the end user to make informed decisions.

4.6 Scalability Considerations

The system is designed to support scalability as the number of EV users increases. Architecture enables distributed data processing and model training using a cloud-based infrastructure.

Large amounts of vehicle telemetry data can be processed using scalable data pipelines. Machine learning models can be updated periodically based on the new usage data available. This makes sure that the system can maintain the accuracy of prediction even when deployed on many vehicles.

In addition, the modular system design enables the independent upgrade of individual components such as data processing or prediction services without involving other modules.

4.7 Security Considerations

Security and data privacy are at the heart of this system, as it deals with the personal vehicle usage patterns of electric vehicle users. There are several layers that ensure that this information is kept secure.

The first is anonymization, which occurs before any analysis is performed on this data. Only the behavioral measurements needed to predict battery life are stored, thereby reducing the overall processing of personally identifiable information. The second is to ensure that this data cannot be easily intercepted while it is in transit. For example, encrypted API connections ensure that when this telemetry is sent from the car to this system, it cannot be easily intercepted by unauthorized parties. The last is to ensure that not everyone can see this information, thereby limiting access to authorized individuals.

These security measures ensure that this system can create useful information without compromising user privacy or data integrity.

5. USER REQUIREMENTS

The section will identify the user requirements of the adaptive battery health prediction system. The identification of the requirements has been done based on the analysis of the purpose of the system, the needs of the targeted stakeholders, and the capabilities of the system that are functional and needed to fulfill the objectives of the study. The section is divided into three subsections, namely stakeholder identification, functional requirements, and non-functional requirements.

5.1 Stakeholder Identification

A stakeholder can be described as any person, group, or organization that has a direct or indirect interest in the system, its outputs, or its operations. Identifying stakeholders is crucial to the requirements engineering process because it ensures that the system will be tailored to meet the real needs of all interested parties. When dealing with the proposed adaptive battery health prediction system, the following three main stakeholder groups can be identified, which are summarized in Table 5.1 below.

Table 5.1: Stakeholder Analysis of the Adaptive Battery Health Prediction System

Stakeholder	Role	Primary Needs	Interaction with System
EV Owner/Driver	Primary end-user	Live battery health status, customized suggestions, predictions of degradation trend.	Uses mobile/web dashboard to look at SOH predictions, get behavioral recommendations, and see trends of battery over time.
EV Manufacturer	System integrator & data provider	Battery usage analytics, warranty, product enhancement.	Uses aggregate levels of battery degradation to enhance battery design and warranty policy.

Vehicle Maintenance & Service Provider	Maintenance decision-maker	Battery health prognostication, predictive maintenance.	Take SOH reports, degradation alerts and plans proactive battery services before failure.

As shown in Table 5.1, the three identified stakeholders, namely the EV owner/driver, the EV manufacturer, and the vehicle maintenance and service provider, have different capacities to interact with the system. The EV owner/driver communicates directly with the dashboard to obtain real-time health data and predictions, while the EV manufacturer and maintenance service provider use the results of the overall system to improve the product and schedule proactive maintenance. Collectively, these stakeholders establish the essence of the user requirements that inform the design and development of the proposed system.

5.2. Functional Requirements

The provided EV battery health prediction system aims to facilitate high-level data collection, predictive analytics, and intelligent decision-making capabilities. The system will be able to collect and store detailed usage information of electric vehicles, as well as information on charging behavior such as charging type (fast or normal) and frequency, charging duration, charging intervals, and state of charge (SOC) after each charge. At the same time, the system will record driving behavior data, which includes vehicle speed, acceleration patterns, braking force, trip length, and energy consumption per kilometer. In addition, battery-specific information, including current trends, voltage levels, temperature, SOC, and internal resistance, will be obtained through the Battery Management System (BMS) to capture accuracy in battery conditions.

The system will then transform the input raw data into meaningful user-specific behavioral characteristics based on the raw data, which include average charging frequency/week, depth of discharge (DoD)/session, rate of fast charging usage,

average trip distance, and acceleration patterns. These will be used to train and deploy machine learning models that can predict the battery state of health (SOH) as a percentage of its original capacity.

The system will also support short-term and long-term predictions of SOH at different time horizons (3 months, 6 months, and 12 months) so that users can know the future trends of battery degradation and make better decisions. In addition, the system approximates the remaining useful life (RUL) of the battery by calculating the remaining time or battery cycles until the SOH reaches a critical level of 80 percent of the initial capacity.

To make it more interpretable, the system will examine the correlation between user actions and battery degradation through explainable machine learning methods. This will enable it to identify key behavioral variables that have a significant impact on battery health, such as aggressive driving, regular fast charging, and deep discharge cycles.

In addition, the system will offer evidence-based and personal recommendations to ensure that battery degradation is minimized and its lifespan is extended. Such recommendations will be rated according to their level of impact (i.e., high, medium, low) so that users can make effective behavioral changes. The system also includes an interactive dashboard, available through web and mobile interfaces, that visualizes the status of the SOH, expected degradation patterns, behaviors, and recommended actions in a clear and easy-to-understand manner.

In addition, the system also allows users to produce and download comprehensive battery health reports that summarize past SOH data, projections, behavior analysis, and recommendations over a given selected time frame. The system also has a history of battery performance and usage behavior, allowing users to see changes over time and gain a clear picture of how their usage patterns are affecting battery performance over time.

5.3 Non-Functional Requirements

Non-functional requirements specify the quality characteristics that ensure the proposed EV battery health prediction system operates effectively, accurately, and reliably under natural conditions. Table 5.2 presents the most urgent requirements of this component.

Table 5.2: Non-Functional Requirements of the Proposed System

Category	Requirement
Performance	The system produces a battery state of health (SOH) within ≤ 10 seconds and can withstand user requests.
Prediction Accuracy	The machine learning model will be able to predict SOH with a mean absolute error (MAE) of no more than 5% to achieve reliable prediction results.
Reliability	The system will be configured to have sustainable operation with minimal failures, as well as basic error handling to avoid interrupted predictions.
Security & Privacy	The system will store and process user data in an encrypted manner, ensuring that personally identifiable information (PII) is not disclosed or misused.
Usability	The system has an easy-to-use interface that enables users to understand SOH predictions and recommendations without any technical knowledge.
Scalability	The system allows for scaling data volumes and number of users without a significant drop in performance.
Availability	The system must ensure continuous access to the latest SOH forecasts and recommendations, including offline access capabilities during temporary network disruptions.

Table 5.2 highlights the most important quality attributes that must be provided for the successful implementation of the proposed system. Performance and accuracy are required to make accurate predictions on a timely basis, while reliability ensures the reliability of the system. Sensitive user data is covered by security and privacy, and usability is aimed at making the system available to non-technical users. Scalability enables the system to support future expansion in the number of users and data and therefore makes it applicable in real-life implementation.

5.4. Use Case Diagram (tentative)

Figure 5.1 shows the Adaptive Battery Health Prediction System using a use case diagram. It illustrates how different users interact with the system and its main functions.

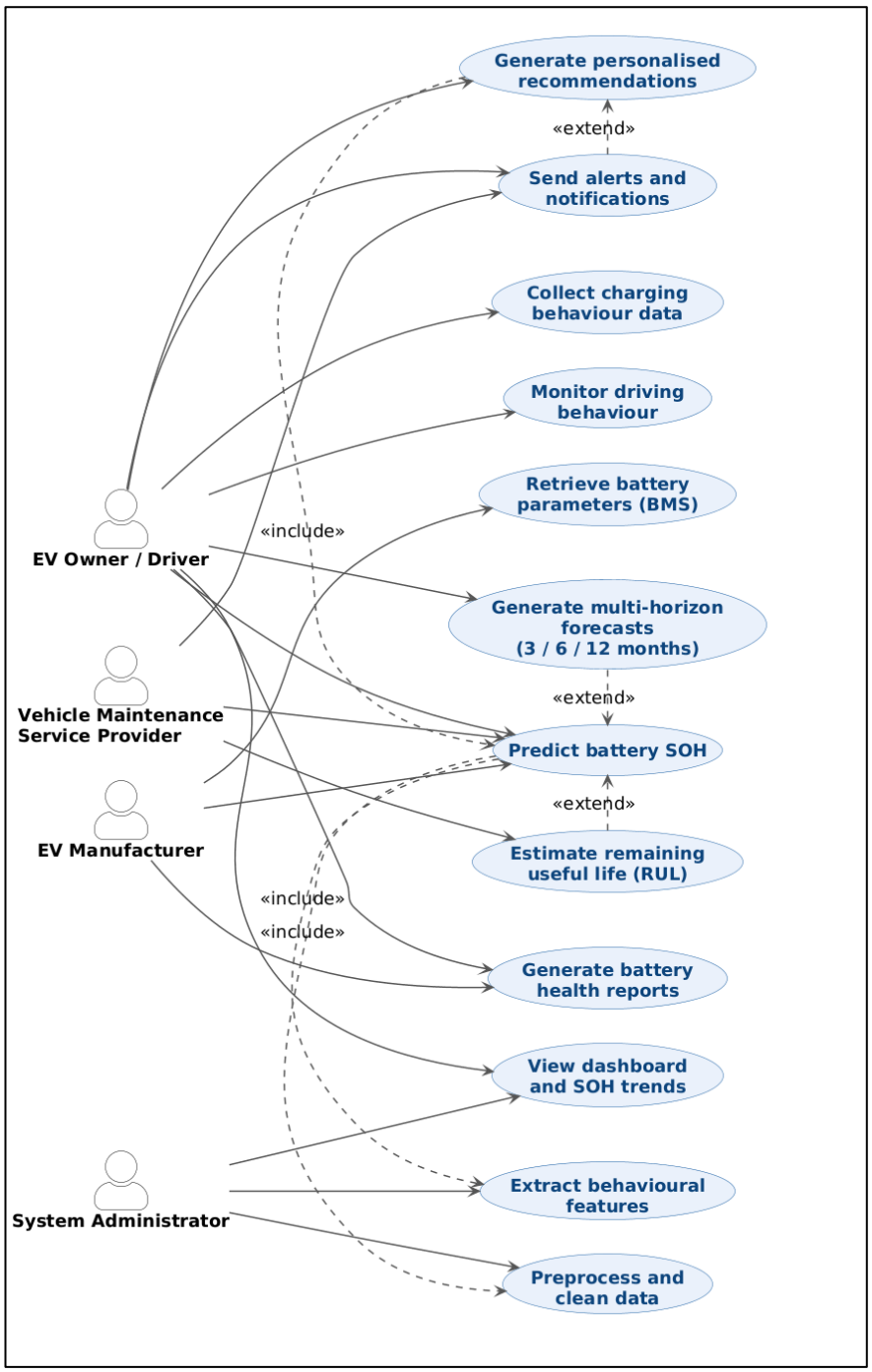


Figure 5.1 Adaptive Battery Health Prediction System

Based on the diagram, the system will serve multiple users, including EV owners, service providers, manufacturers, and administrators. It includes key roles such as data collection, battery health prediction (SOH), remaining useful life prediction, and producing recommendations and reports. The diagram shows how the system analyzes the data and provides valuable insights to improve battery performance.

5.5 User Stories

EV Owner / Driver

US-01: Glance at Battery Health

Being an owner/driver of an EV, I would like to check the percentage of my battery's State of Health (SOH) on my mobile or web dashboard to be able to know how healthy my battery is now and make correct decisions regarding the use of my vehicle.

US-02: Accept Degradation Trend Forecasts

As an EV owner/driver, I would like to know the short-term and long-term projections of the way my battery health is going to change in the coming 3, 6 and 12 months to plan my battery maintenance or replacement before my capacity suddenly reduces.

US-03: Receive Personalized Recommendations

Being an EV owner/driver, I would like to be provided with personalized and evidence-based advice on my charging and driving patterns, thus being able to make certain decisions to reduce the pace of battery degradation and prolong my battery life as much as possible.

EV Manufacturer

US-04: Get Aggregated Fleet Battery Data

Being an EV manufacturer, I would like to access aggregated and anonymized data of battery degradation on a population of real-world vehicles so that I can understand some general trends of degradation and apply that knowledge to subsequent battery designs and vehicle performance.

US-05: Justify Warranty Management Decisions

Since I am an EV manufacturer, I would like to have access to SOH data and forecasts of degradation trends of vehicles still under warranty to ensure I can make decisions

based on data and determine the batteries that degrade faster than I had expected considering specification standards.

US-06: Recognize Degradation Patterns of High-Risk

Being an EV manufacturer, I would like to be notified when unusual or accelerated battery degradation patterns are observed within the vehicle fleet, such that I would be able to research possible manufacturing flaws or design shortcomings before it progresses into a large-scale product reliability concern.

Vehicle Maintenance & Service Provider

US-07: Select SOH reports on a case-by-case basis of vehicles

Being a vehicle maintenance service provider, I would like to have a detailed state of health report of any single vehicle presented to me because I would like to easily and effectively determine the state of the battery and deliver a valid diagnosis to the customer without relying on the built-in display on the vehicle alone.

US-08: Predictive Maintenance Alerts Received

Since I am a vehicle maintenance service provider, I would be interested to have early warning notifications once the battery SOH of a vehicle is predicted to fall below the critical end-of-life value of 80 percent within a specific time frame in the future so that I can be proactive and schedule a battery check or change before the customer suffers a sudden failure.

US-09: Full Degradation Trend History

Because I am a vehicle maintenance service provider, I would like to see the whole battery degradation history of a vehicle (the previous SOH readings, charging behavior pattern, driving intensity data, etc.), which will allow me to diagnose the actual cause of battery degradation precisely and provide the customer with the specific and informed maintenance recommendation.

6. COMMERCIALIZATION PLAN

6.1. Target market and customer persona

The proposed adaptive battery health prediction system will focus on three main customer segments. Initially, the primary market for individual EV owners is likely to be urban and tech-savvy users who can use an electric car as a means of daily commuting and are concerned about battery life and resale value. Rising fuel prices and the rise of electric mobility have led to rapid growth of this segment in Sri Lanka. Secondly, EV manufacturers and automotive original equipment manufacturers (OEMs) are an important business-to-business (B2B) segment. Such stakeholders need to have on-the-ground battery performance analytics to better design products, optimize warranty strategies, and increase competitiveness.

Thirdly, another segment that is becoming important is vehicle maintenance and service providers, which can be in the form of authorized service centers, independent garages, and fleet management companies. Such users benefit from predictive maintenance knowledge that reduces vehicle downtime and increases service efficiency.

Table 6.1 includes the key characteristics of these customer groups.

Table 6.1: Target Market Segments and Personas

Segment	Characteristics	Needs
Individual EV Owners	Urban, 25–50 years old, tech-savvy, daily EV users	Understanding battery health, cost savings, recommendations
EV Manufacturers (OEMs)	Automotive companies and engineers	Fleet level analytics, product improvement
Service Providers & Fleets	Garages, service centers, fleet operators	Predictive maintenance, reduced downtime

Table 6.1 breaks down the main customer groups and highlights their key characteristics and needs. This differentiation will allow the system to focus more on delivering value to each group and to the individual user and stakeholder of the enterprise.

6.2. Value proposition

The system can be broken down into a unique value proposition as it provides personalized, predictive, and actionable battery health intelligence that cannot be provided in traditional battery management systems (BMS). Instead of showing simple metrics like state of charge (SOC), the proposed system predicts battery degradation (3, 6, and 12 months) based on real-world behavioral data and translates them into realistic recommendations, unlike traditional systems.

For EV users, the system reduces the chances of battery failure and unnecessary battery replacement costs (LKR 1.5M-3M) and improves battery health confidence. For manufacturers, it provides access to real-world analytics that can improve product design as well as reduce product liability. For service providers, it helps in predictive maintenance, thereby providing greater efficiency and customer satisfaction.

6.3. Revenue model

This system follows a multi-stream revenue model targeting different stakeholder groups. The main revenue streams are summarized in Table 6.2.

Table 6.2: Revenue Model

Revenue Stream	Description	Pricing
Subscription (B2C)	Individual EV users (Freemium + Premium)	LKR 1,200/month or LKR 10,000/year
B2B Licensing	OEM and vehicle line operators (API/white label)	LKR 5M – 30M/year

Service Provider Subscription	Garages and service centers	Starting from LKR 8,000/month

Table 6.2 presents the system's diversified revenue streams. The combination of B2C subscriptions and B2B licenses provides the ability to scale across multiple market segments while ensuring financial sustainability.

6.4. Pricing strategy

The proposed system will be based on a value-based and penetration pricing strategy that ensures that it is affordable, fast, and can be adopted in the long term. The prices will be adjusted according to the economic benefit provided to the end users, especially in terms of minimizing the battery depletion costs and eliminating the high cost incurred by the user when changing the battery. Moreover, different pricing tiers have been introduced to support individual users, enterprise clients, and service providers. The comprehensive pricing schedule is given in Table 6.3.

Table 6.3: Pricing Strategy

Customer Segment	Plan Type	Features	Pricing
Individual EV Users	Free (Freemium)	Initial SOH observation, limited 3-month forecast	LKR 0
Individual EV Users	Premium	Full 3, 6, 12-month forecasts, personal recommendations, alerts, reports	LKR 1,200/month or LKR 10,000/year
Individual EV Users (Launch Offer)	Discounted Plan	Same as premium (limited time)	LKR 800/month

EV Manufacturers / OEMs	B2B Licensing	API access, fleet analytics, white label integration	LKR 5M – 30M/year
Service Providers	Subscription	Vehicle diagnostics, predictive maintenance understanding	Starting from LKR 8,000/month
Fleet Operators	Enterprise Plan	Bulk vehicle monitoring, analytics dashboard, priority support	Custom pricing (based on fleet size)

Table 6.3 summarizes the pricing policy of the different segments of the customers. The freemium is offered to individual users as an introduction and motivates them to use the product, while the premium version is one that offers full functionality, long-term forecasts, and customized recommendations. The initial launch will be done at a discounted price in the short term to ensure rapid penetration of the product in the market. For enterprise customers such as OEMs and fleet operators, it is flexible and depends on size and features. This tiered pricing is affordable for the average user while still getting the most out of business clients.

6.5. Competitive advantage

This system offers several sustainable competitive advantages, which are summarized in Table 6.4.

Table 6.3: Competitive Advantages

Advantage	Description
Behavior-driven prediction	Real-world driving and changing behavior are used for personalized predictions.
Adaptive learning	The model continuously improves with new data (data network effect)

Localization	Optimized for Sri Lankan conditions (temperature, driving patterns).
Integrated platform	Combines predictions + analytics + recommendations
Multi-stakeholder support	Serving users, OEMs, and service providers from one system.

Table 6.3 highlights the unique strengths of the proposed system. These advantages differentiate the system from traditional BMS and existing diagnostic tools and create strong barriers to entry for competitors.

6.6. Intellectual property considerations

The system contains several elements that can be considered intellectual property (IP). The core innovation is that it combines charging behavior, driving patterns, and battery parameters to provide personalized SOH prediction and recommendations.

This algorithm can be patented using the Intellectual Property Act of Sri Lanka, an intellectual property law. In addition, the name and branding of the system must be registered as trademarks to prevent imitation in the market. The source code, training models, and database structures are covered by copyright protection.

Moreover, data sharing agreements with external parties, including non-disclosure agreements (NDAs) and license terms, should be in place to avoid any disclosure of proprietary data and to ensure that the data is not misused. The combination of these IP strategies ensures long-term competitive advantage and sustainability during commercialization.

7. BUDGET

This section describes the estimated budget to be used for the development and implementation of the proposed adaptive battery health prediction system. The budget covers all the costs required for hardware, cloud services, software, and human effort costs as shown in Table 7.1 below.

Table 7.1: Budget Estimate

Category	Item Description	Estimated Cost (LKR)	Justification
Hardware	Laptop/PC upgrade (RAM/SSD)	25,000	Required for running ML models and preprocessing EV data efficiently.
	External storage (1TB SSD)	10,000	Secure storage for datasets and backups.
Cloud Services	Cloud GPU instance (Google Colab Pro / AWS EC2 for 3 months)	15,000	For training ML models on large datasets efficiently without overloading local hardware.
Software	Python IDE + libraries (PyCharm Pro, optional)	10,000	Professional IDE for coding, debugging, and ML model development.
	Data visualization software (Tableau or equivalent student license)	10,000	To generate interactive dashboards and graphs for battery health insights.

Human Effort Cost	Transportation to company / site visits	20,000	Travel expenses for data collection, sensor installation, and behavioral study.
Total		90,000	

The total project estimated budget as shown in Table 7.1 will be Rs. 90,000. The largest budget will be spent on hardware resources, which are essential in implementing machine learning models and securing data management. Cloud services will support model training, deployment, and scaling, while software costs will be reduced by using open source and student licenses. Human effort costs include field work required for data collection and conducting behavioral research. This budget plan has been generated to successfully implement, test, and demonstrate the proposed system within the approved financial limit of an academic research project.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1. Task Breakdown Per Member

To ensure that the entire project is developed systematically and the allocation of tasks is effectively organized, the entire project is divided into several components and work packages. The hierarchical breakdown of the EV Intelligent Monitoring and Forecasting System into manageable tasks and subtasks is shown through the Work Breakdown Structure (WBS) in Figure 8.1.

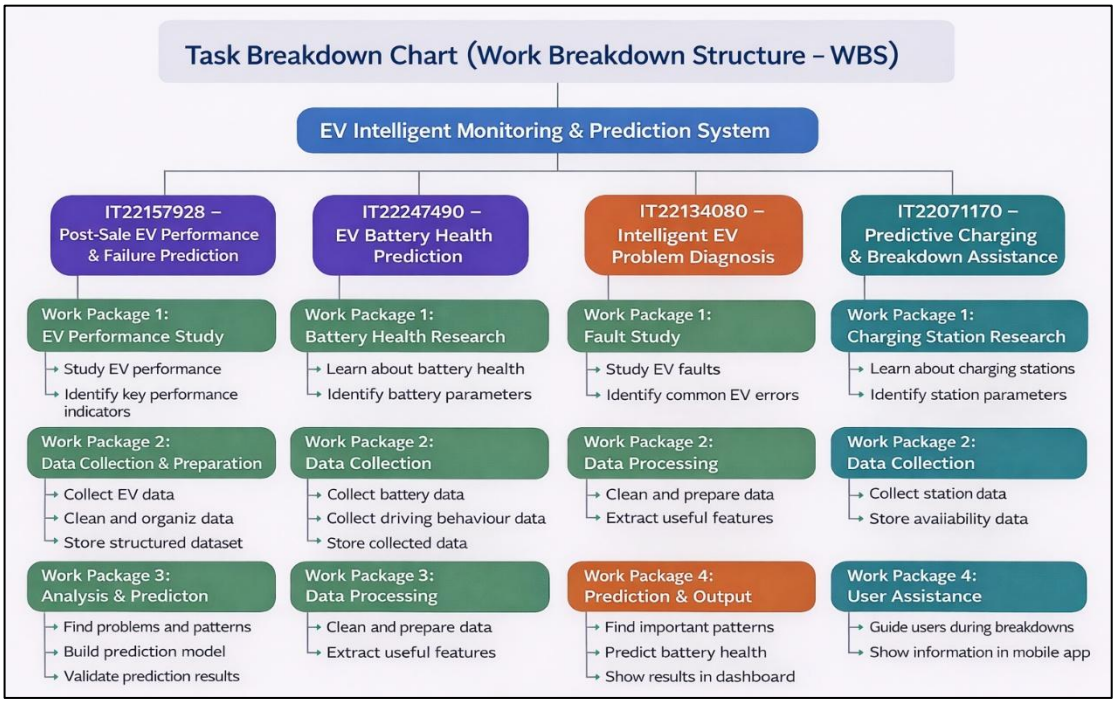


Figure 8.1: Work Breakdown Structure of EV Prediction System

As shown in Figure 8.1, the entire system is divided into four major parts, namely, aftermarket EV performance and failure prediction, EV battery health prediction, intelligent EV problem diagnosis and predictive charging, and breakdown support. The components are further divided into different work packages to ensure a clear development flow and distribution of responsibilities.

Focusing on the EV battery health prediction component, the work is divided into three main phases, namely battery health research, data collection, and data processing.

These phases include determining critical battery parameters, collecting both battery and behavioral data, and data cleaning and feature extraction. This formal model ensures that the prediction model is created with properly prepared and meaningful information.

At the same time, the other parts have a logical flow of research and data collection for their processing, prediction, and user support. Such a hierarchical decomposition helps manage project complexity, allocate tasks, and develop all components of the system in a coordinated and systematic manner.

8.2. Balanced Workload Distribution

The table below shows the workload shared by the team members working on the project. It describes how responsibilities are assigned, work is shared, and collaboration is maintained to ensure the successful development of the system.

Table 8.2 Workload Distribution

Aspect	Description
Task Allocation	Each team member is responsible for one major system component.
Work Distribution	All members contribute equally through research, system design, model development, implementation, and testing.
Collaboration	Team members collaborate during system integration to combine all components into one complete EV Intelligent Decision Support System.
Project Outcome	The modular development approach ensures balanced workload, efficient development, and a fully integrated intelligent EV decision support platform.

As shown in Table 8.2, the project is organized in terms of systematic workload distribution. Team members will be assigned to certain parts of the system, which means that tasks and responsibilities are clearly distributed. Meanwhile, all members are engaged in key activities such as research, system design, model development, implementation, and testing, which encourages equal contributions. Teamwork is relevant during the system integration phase, where separate components are integrated into a fully functional EV intelligent decision support system. This integrated or coordinated style helps keep the development process efficient and balances the team's workload with the result.

9. GANTT CHART

Figure 9.1 is a layout of the Research Project (RP) schedule for 2026. It depicts the significant phases of the project and important project deliverables throughout the year. It is a visual representation of the progress of the project from the initial stages to the final stages.

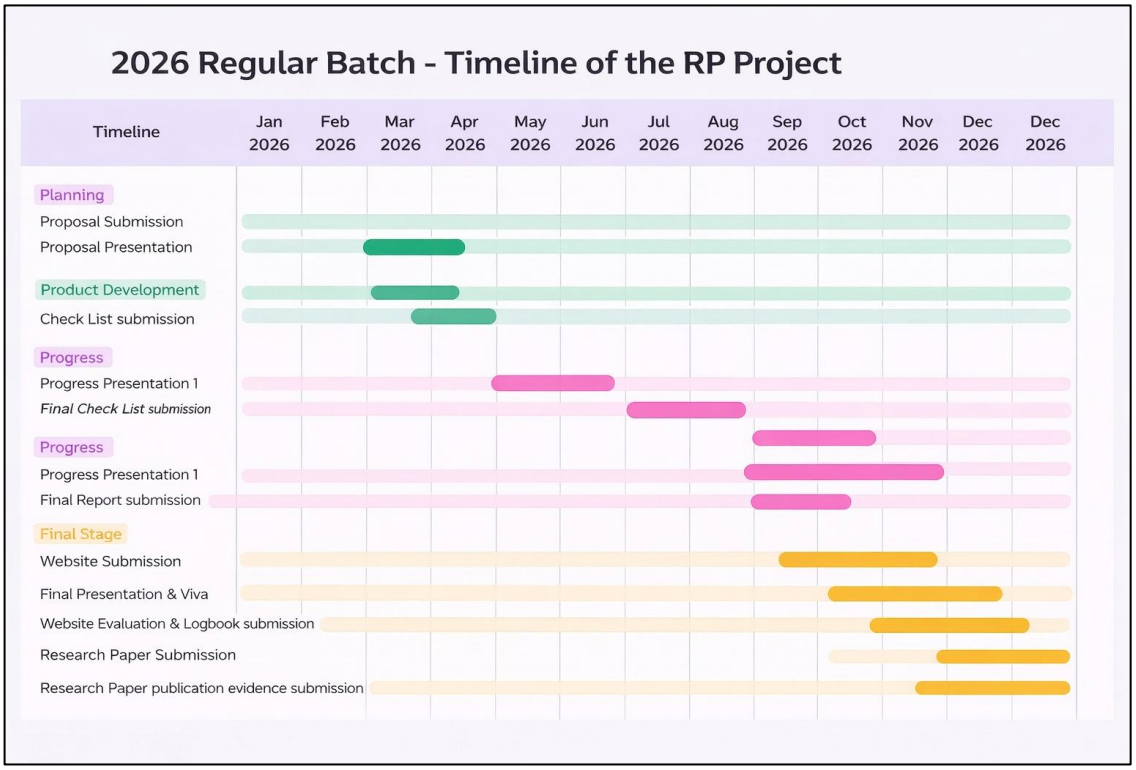


Figure 9.1: Gantt Chart

The project begins in January and February with planning activities such as proposal submission and presentation. Development begins in February, and this continues with other key activities such as submission of checklists and regular progress reviews. In the final phase of the project from October to December, the main activities include delivering key deliverables, including website submission, final presentation, submission of research papers, and preparation of publication evidence. The project is well planned and outlined for smooth progress and completion.

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11. APPENDICES

11.1. Survey Instruments

The survey tool below was aimed at gathering primary data on EV users in terms of battery usage patterns, knowledge on battery degradation, and readiness to use a predictive battery health system. The survey has been administered to EV owners and drivers in Sri Lanka in the context of the first stage of the research, which is the fieldwork of requirements collection and feasibility analysis.

Survey Title: EV Battery Health Awareness and Usage Behavior Survey.
Guidelines: You are expected to give honest answers to all questions. This survey will be anonymous, and the information gathered will not be utilized in any other way other than academic research. The amount of time required to complete it should be around 10 minutes.

Part A - Profile of the participant.

A1. What is your age group?

☐18 - 25

☐ 26 - 35

☐ 36 - 45

☐ 46 - 55

☐ 56 and above

A2. What is your occupation?

☐ Student

☐ Government employee

☐ Private sector employees

☐ Business owner / Self-employed.

☐ Other: _____

A3. What is the length of ownership or operation of an electric vehicle?

☐ Less than 6 months

☐ 6 months - 1 year

☐ 1 - 3 years

☐ More than 3 years

A4. What is the kind of electric car that you own and operate?

☐ Fully Electric Vehicle (BEV)

☐ A plug-in hybrid electric car (PHEV)

☐ Hybrid Electric Vehicle (HEV)

Part B - Charging Behavior.

B1. What is your charge rate of EV per week?

☐ Once

☐ 2 - 3 times

☐ 4 - 5 times

☐ Daily

B2. Which category of charging do you mostly use?

☐ Home charger standard (slow) charging.

☐ Quick charge (public station)

☐ Both equally

B3. On which battery level do you usually start charging?

☐ Below 20%

☐ 20% - 40%

☐ 40% - 60%

☐ Above 60%

B4. Do you put your battery to a full charge on a regular basis?

☐ Yes, always.

☐ Sometimes

☐ Rarely

☐ Never

B5. Have you had any acute or unaccounted loss of driving range?

☐ Yes

☐ No

☐ Not sure

Part C—Driving Behavior

C1. How many miles do you drive every day?

☐ Less than 20 km

☐ 20 - 50 km

☐ 50 - 100 km

☐ More than 100 km

C2. What is your driving personality?

☐ Very smooth and gentle.

☐ Moderate

☐ Aggressive and has frequent hard acceleration.

☐ Deviated depending on circumstance.

C3. Do you often drive at high speed (more than 100 km/h)?

☐ Yes, regularly.

☐ Occasionally

☐ Rarely

☐ Never

C4. How frequently do you travel on trips less than 5 km?

☐ Daily

☐ Several times a week

☐ Occasionally

☒ Never

Part D — Battery Health Awareness

D1. Do you know what Battery State of Health (SOH) is?

☐ Yes, I fully understand it.

☐ Since I have heard of it but do not know what it is.

☐ No, I have never heard of it.

D2. Does your car show any battery health details other than the battery charge percentage?

☐ Yes

☐ No

☐ I am not sure.

D3. Have you provided service to have your EV battery checked or tested by a service provider?

☐ Yes

☐ No

D4. What is your worry regarding the long-term worsening of your EV battery?

☐ Very concerned

☐ Moderately concerned

☐ Slightly concerned

☐ Not concerned at all.

D5. Do you think that your charge/drive cycles influence your battery life?

☐ Yes

☐ Possibly

☐ No

☐ I am not sure.

Part E- System Acceptance and Willingness to Use

E1. Do you use a mobile or a web application that forecasts your battery life and offers personalized suggestions to maximize battery life?

☐ Yes.

☐ Probably yes.

☐ Not sure.

☐ Probably not.

☐ Definitely not.

E2. What do you consider to be the most useful aspects of a battery health prediction system? (Select all that apply)

☐ SOH percent real-time display.

☐ 3, 6, and 12 months' degradation trend predictions.

☐ Individualized charging prescriptions.

☐ Individual driving suggestions.

☐ Notifications on the battery health lowering the threshold.

☐ The battery health reports can be downloaded.

☐ Historical view of data on batteries and behavior.

E3. To what extent would you pay a premium for a battery health predictive service monthly?

☐ I would not be paid free only.

☐ LKR 500 - 1,000 per month

☐ LKR 1,000 - 1,500 per month

☐ LKR 1,500 - 2,500 per month

☐ More than LKR 2,500 per month

E4. What is your level of concern with data privacy in a battery monitoring application?

☐ Extremely important

☐ Very important

☐ Moderately important

☐ Not important

11.2. Detailed Diagrams

The general architecture of the proposed EV research and intelligence system is shown in Figure 11.1. The system integrates four interconnected modules aimed at providing end-to-end intelligence to stakeholders in electric vehicles, running on a single core of machine learning, deep learning, and NLP-based analytics.



Figure 11.1: Overall System Architecture

The system is organized into four large groups, as shown in Figure 11.1, with each group having different user groups and system functions. **Component 1 (EV Sales Company)** provides after-sales EV performance intelligence, collecting usage data and cloud analytics to determine failure patterns, creating battery reports, and offering dealer dashboards with warranty information. Example **Component 2 (EV User - Adaptive Battery Health Prediction)** uses ML models to predict battery health over 3-, 6-, and 12-month trend windows using data on charging and driving behavior and provide degradation cues through user feedback loops. **Component 3 (EV User EV Problem Diagnosis and Guidance)** uses an LSTM-based autoencoder to detect anomalies, a machine learning classifier to classify anomalies as critical, moderate, or

safe, and NLP rules to create contextual dos and don'ts and emergency alerts. **Component 4 (EV User Predictive Charging/Assistance)** consists of the following features: proactive charging predictions, real-time location, crash notifications including SOS, optimized route navigation, and emergency rescue guidance. All these features are elements of a holistic, AI-engineered ecosystem that improves the reliability of EVs, their safety for users, and their performance.

The machine learning pipeline used to predict EV battery health is systematic and structured, as shown in Figure 11.2. The workflow ensures that predictive model development is performed on clean, pure data and that the model is thoroughly tested before deployment so that battery health predictions are accurate and reliable.

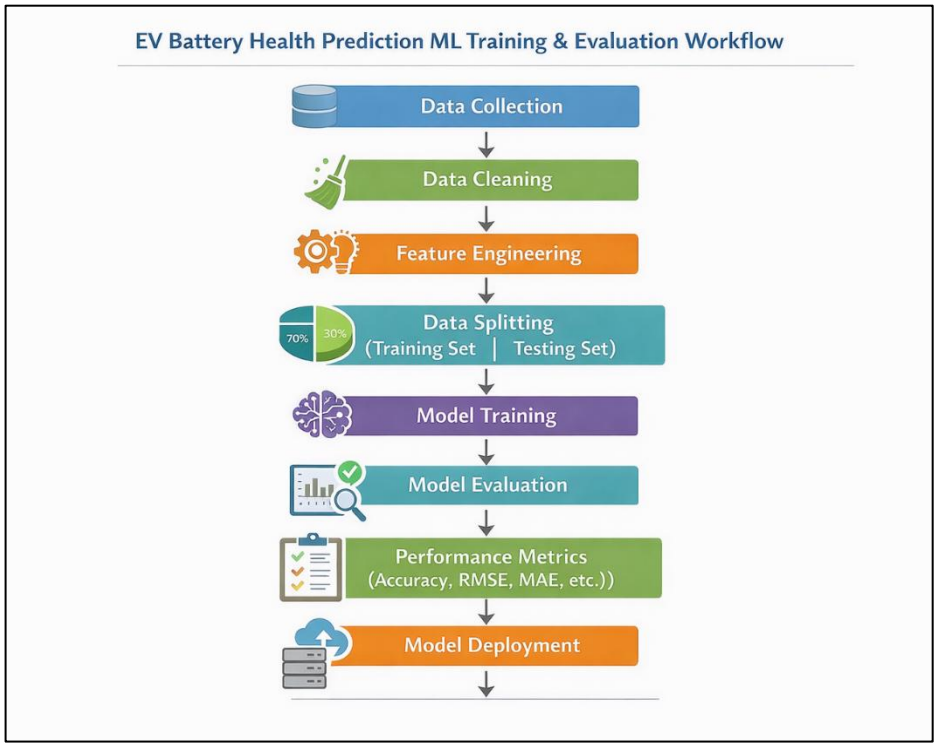


Figure 11.2: Machine Learning Model Training and Evaluation Workflow

This figure shows that the ML training and evaluation workflow has eight steps that follow a series of steps. It starts with data collection, where raw parameters of the EV battery, including charging cycles, voltage, temperature, and usage patterns, are collected according to relevant sources. This phase is followed by data cleaning, which

involves handling missing values, removing noise, and correcting inconsistencies to ensure data quality. The feature engineering phase involves deriving meaningful features that can be translated into better predictions from the model. In the data-splitting step, the data set is divided into a 70 percent training set and a 30 percent test set so that the evaluation is unbiased. In terms of model training, machine learning algorithms are used to process the training data to learn battery degradation patterns and health trends. The trained model is then evaluated through model evaluation, where its performance is compared with unknown test data. Key performance indicators – such as accuracy, root mean square error (RMSE), mean absolute error (MAE), and other appropriate measures – are calculated to determine the usefulness of the model. Finally, when it is satisfactorily tested, the model is then taken to model deployment, where it is incorporated into the existing system that produces end-user predictions of battery health in real time. This is a comprehensive end-to-end workflow, which ensures that the deployed model is scientifically valid and practically sound.

11.3 Preliminary UI Mockups

To test the functionality of the adaptive battery health prediction component design, a prototype user interface was created to display the battery real-time status and future health predictions. This prototype dashboard presents the first screens as shown in Figures 11.3 and 11.4.

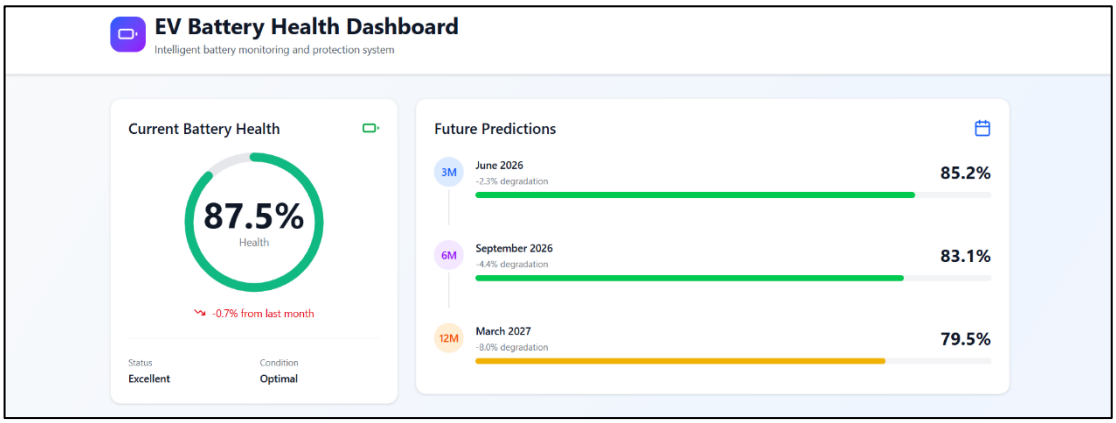


Figure 11.3: Prototype User Interfaces 1

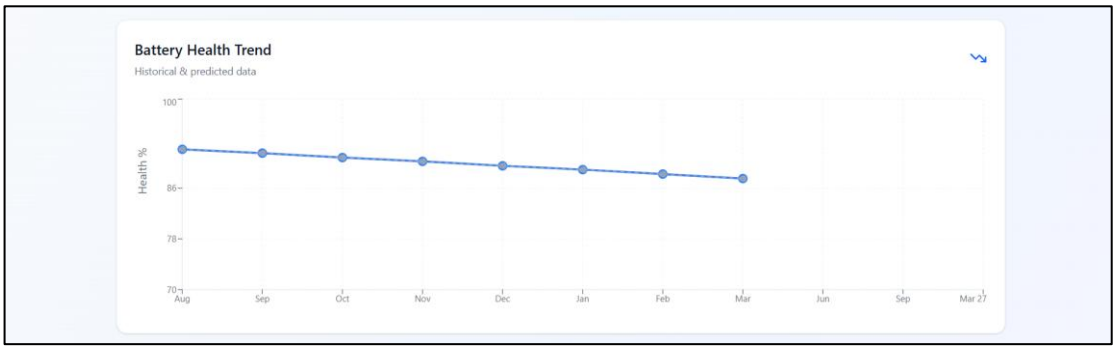


Figure 11.4: Prototype User Interfaces 2

As can be seen in Figures 11.3 and 11.4, the EV Battery Health Dashboard is a simplified dashboard that allows users to get a clear and concise overview of the status of their vehicle's battery as well as predictive information. The key dashboard (shown in Figure 11.3) displays the status of the battery health in the form of a visual gauge and a list of future health predictions in an organized, easy-to-read format. Figure 11.4 complements this by showing the battery's past and future performance over time, allowing users to graphically see changes in battery health over time. Together, these two screens form the system's central monitoring interface, allowing EV users to monitor the health of their battery briefly.

Additional prototype screens were developed to provide the user with more behavioral analysis and practical advice, as shown in Figures 11.5 and 11.6.

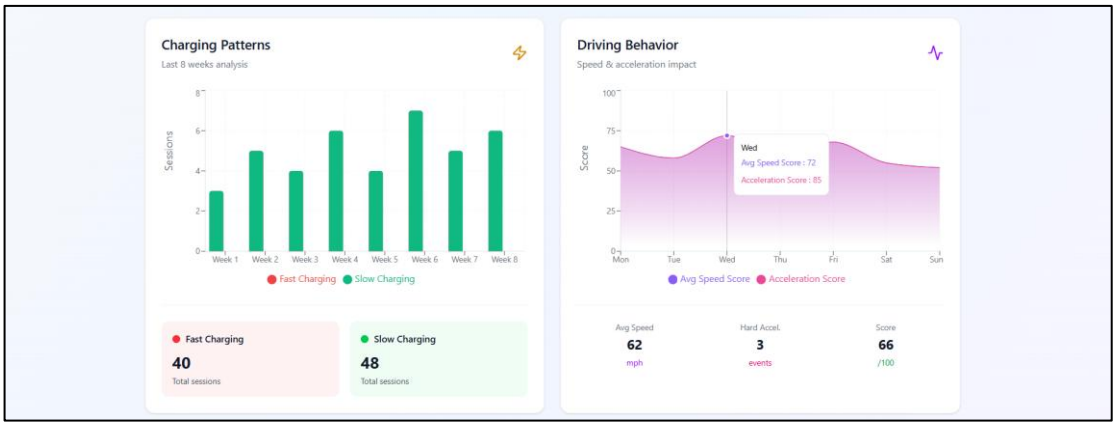


Figure 11.5: Prototype User Interfaces 3

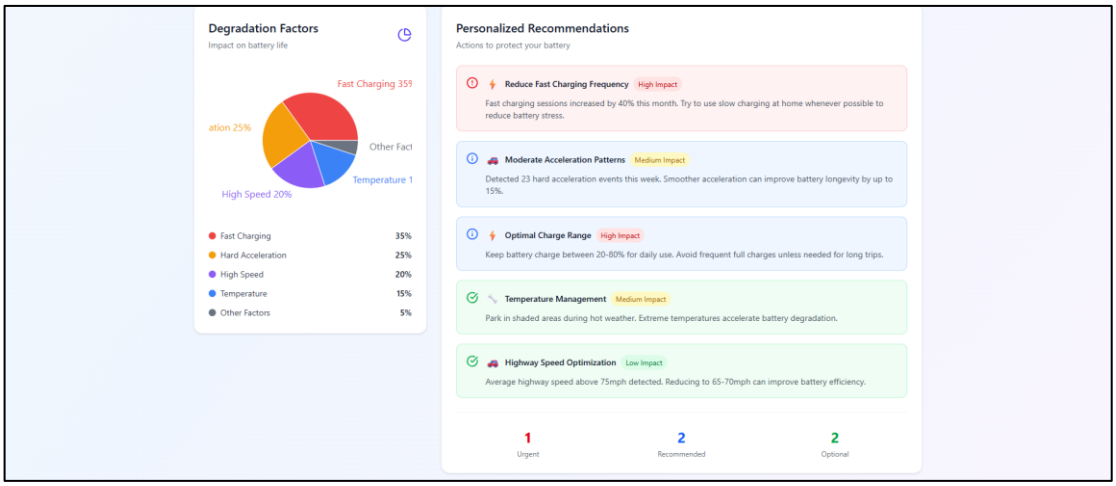
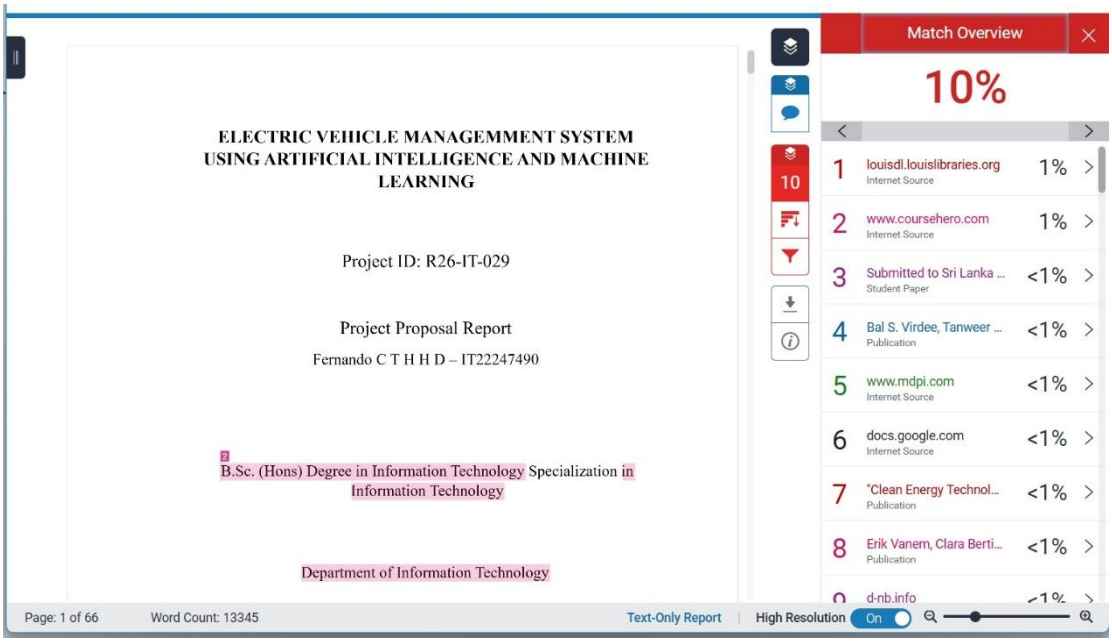


Figure 11.6: Prototype User Interfaces 3

The prototype interface, as shown in Figures 11.5 and 11.6, goes beyond simple health monitoring and provides detailed usage analytics and customized suggestions. Figure 11.5 shows a dual-pane display of charging patterns and driving behavior visualizations, where the user can understand daily habits and how they affect battery performance with the help of interactive charts and summarized statistics. Figure 11.6 presents a degradation factors panel and recommended actions section, where the system identifies the most influential factors in battery degradation and provides prioritized user-friendly recommendations to reduce further degradation. Such a mix of diagnostic knowledge and practical advice demonstrates the system's goal to enable EV users to make unbiased decisions that maximize battery capacity and overall vehicle performance.


ELECTRIC VEHICLE MANAGEMMENT SYSTEM USING ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
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